

1st Exam 2025/2026

Save the tasks in the file `naloge.py`.

Task 1: Read the data

The data about programming students are stored in a `csv` file. Assume that the file has a format consistent with the example below. A comma (,) is used as the delimiter.

Example content of the file `python5.csv`:

```
id,starost,drzava,izkusnje,tedni,ure,naloge,videi,samoocena,tocke,opravljeno
600,43,Indonesia,Intermediate,15,8.2,62,30,10,32.5,0
1200,52,UK,Beginner,2,4.4,62,36,5,32.1,0
1800,16,UK,Beginner,7,10.1,54,40,7,41.6,0
2400,21,Indonesia,Beginner,15,11.2,53,29,3,34.9,0
3000,36,Pakistan,Advanced,12,3.6,57,48,7,59.8,0
```

Write a function `preberi_podatke` that takes the file name as an argument and returns a list of tuples of the form

`(starost, drzava, izkusnje, tedni, ure, naloge, videi, ocena).`

The following hold:

- `starost` represents the student's age, stored as `int`,
- `drzava` represents the country the student comes from, stored as `str`,
- `izkusnje` represents the student's prior experience, stored as `str`,
- `tedni` represents the number of weeks the student attended the course, stored as `int`,
- `ure` represents the number of hours per week the student spent on the course, stored as `float` and rounded to 1 decimal place,
- `naloge` represents the number of solved assignments, stored as `int`,
- `videi` represents the number of watched course videos, stored as `int`, and
- `ocena` represents the final exam grade, stored as `int`.

The value of the variable `ocena` is obtained by assigning a numerical grade on the interval 5–10 to the variable `tocke` from the file according to the following criteria:

- fewer than 50 points corresponds to grade 5,
- from 50 (inclusive) to 60 points corresponds to grade 6,
- from 60 (inclusive) to 70 points corresponds to grade 7,
- from 70 (inclusive) to 80 points corresponds to grade 8,
- from 80 (inclusive) to 90 points corresponds to grade 9, and

- from 90 (inclusive) to 100 (inclusive) corresponds to grade 10.

When creating the list of tuples, skip the row containing the column names.

If the value 'Other' appears in the variable `country`, do not include the data from that row in the list of tuples.

Example of a correct function execution:

```
>>> preberi_podatke('data/python5.csv')
[(43, 'Indonesia', 'Intermediate', 15, 8.2, 62, 30, 5),
 (52, 'UK', 'Beginner', 2, 4.4, 62, 36, 5),
 (16, 'UK', 'Beginner', 7, 10.1, 54, 40, 5),
 (21, 'Indonesia', 'Beginner', 15, 11.2, 53, 29, 5),
 (36, 'Pakistan', 'Advanced', 12, 3.6, 57, 48, 6)]
```

Task 2: Grade dictionary

Write a function `v_slovar` that takes as an argument a list of tuples as returned by Task 1 (i.e. a list of tuples of the form

`(age, country, experience, weeks, hours, assignments, videos, grade)`).

The function should return a dictionary where the keys are grades and the values are lists of tuples of the form `(age, country, experience, weeks, hours, assignments, videos)`.

Example of a correct function execution:

```
>>> podatki = [(43, 'Indonesia', 'Intermediate', 15, 8.2, 62, 30, 5),
               (52, 'UK', 'Beginner', 2, 4.4, 62, 36, 5),
               (16, 'UK', 'Beginner', 7, 10.1, 54, 40, 5),
               (21, 'Indonesia', 'Beginner', 15, 11.2, 53, 29, 5),
               (36, 'Pakistan', 'Advanced', 12, 3.6, 57, 48, 6)]
>>> v_slovar(podatki)
{5: [(43, 'Indonesia', 'Intermediate', 15, 8.2, 62, 30),
      (52, 'UK', 'Beginner', 2, 4.4, 62, 36),
      (16, 'UK', 'Beginner', 7, 10.1, 54, 40),
      (21, 'Indonesia', 'Beginner', 15, 11.2, 53, 29)],
 6: [(36, 'Pakistan', 'Advanced', 12, 3.6, 57, 48)]}
```

Task 3: Grade statistics

Write a function `statistika` that takes as argument a dictionary as returned by Task 2 (i.e. a dictionary where the keys are grades and the values are lists of tuples of the form (age, country, experience, weeks, hours, assignments, videos)). The function should return a dictionary where the keys are grades and the values are the average number of hours spent programming. The averages should be rounded to two decimal places.

Example of a correct function execution:

```
>>> slovar = {5: [(43, 'Indonesia', 'Intermediate', 15, 8.2, 62, 30),  
                (52, 'UK', 'Beginner', 2, 4.4, 62, 36),  
                (16, 'UK', 'Beginner', 7, 10.1, 54, 40),  
                (21, 'Indonesia', 'Beginner', 15, 11.2, 53, 29)],  
             6: [(36, 'Pakistan', 'Advanced', 12, 3.6, 57, 48)]}  
>>> statistika(slovar)  
{5: 8.47, 6: 3.6}
```

Task 4: By experience

Write a function `po_izkusnjah` that takes as arguments a list of tuples of the form

`(age, country, experience, weeks, hours, assignments, videos, grade)`

as returned by Task 1, and a string representing the experience level of the participants.

The function should return the name of the country (`str`) from which the largest number of participants with the given experience level comes. You may assume that in each case there is always at most one such country.

If the function is given an empty list, or if there are no participants with the given experience level in the list, the function should return an empty string.

Hint: first prepare a dictionary where the keys are countries and the values are the numbers of participants with the given experience level coming from that country.

Example of a correct function execution:

```
>>> podatki = [(52, 'UK', 'Beginner', 2, 4.4, 62, 36, 5),
               (16, 'UK', 'Beginner', 7, 10.1, 54, 40, 5),
               (21, 'Indonesia', 'Beginner', 15, 11.2, 53, 29, 5),
               (36, 'Pakistan', 'Advanced', 12, 3.6, 57, 48, 6)]
>>> po_izkusnjah(podatki, 'Beginner')
'UK'
>>> po_izkusnjah(podatki, 'Zacetnik')
''
```